PHYDSC403T — Basic Quantum Mechanics

Unit 3: Applications of Schrodinger Equation

7th Semester, Assam University (FYUG) — Haflong Government College

Paper: PHYDSC403T, Full Marks 100 [ESE 70 + CCA 30] | Unit 3: 13 lectures

Notes 13 of 20 — drafted by gemini-3.6-flash (Kaggle Model Proxy) from the unit syllabus only, without external sources, and checked against the syllabus.

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Unit syllabus (as in the syllabus file)

1. Free particle in one dimensional infinite potential well and calculation of its eigen values and normalized eigen functions.
2. Particle in one dimensional and three-dimensional box concept of degeneracy.
3. Calculation for transmission and reflection coefficient for particle encountering step potential.
4. Particle inside finite rectangular potential barrier - Phenomenon of quantum tunneling.

Top 2 high-mark topics in this unit

1. Particle in a 1-D infinite well and in a 1-D / 3-D box, energy eigenvalues and normalized eigenfunctions, and degeneracy — 4 of 4 AU questions in this unit are on the box, including a full solve in Sec B.
2. Step potential transmission and reflection coefficients, and finite rectangular barrier / quantum tunnelling — 13 lectures, the heaviest unit, full derivation, and an explicit AU gap, a standard long question in the Calicut essay pattern.

Contents

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2. Particle in a Three-Dimensional Box and Degeneracy
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5. Questions — Short form, then Long form (4 AU questions)
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1. Particle in a One-Dimensional Infinite Potential Well

Consider a particle of mass m constrained to move along the x-axis between two impenetrable boundaries located at $x = 0$ and $x = a$. This system is known as a particle in a one-dimensional infinite potential well or a one-dimensional box. The potential energy function $V(x)$ defining this system is given by:

$$V(x) = \begin{cases} 0 & \text{for } 0 < x < a \\ \infty & \text{for } x \leq 0 \text{ and } x \geq a \end{cases}$$

Because the potential energy is infinite outside the region $0 < x < a$, the particle cannot penetrate into the regions $x \leq 0$ or $x \geq a$. Consequently, the probability of finding the particle outside the well is zero, and the wave function $\psi(x)$ must vanish identically in those regions:

$$\psi(x) = 0 \quad \text{for } x \leq 0 \text{ and } x \geq a$$

Setting up the Schrödinger Equation

Inside the well ($0 < x < a$), the potential energy is zero, $V(x) = 0$. The time-independent Schrödinger equation for the spatial wave function $\psi(x)$ in one dimension is:

$$-\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2} + V(x)\psi(x) = E\psi(x)$$

Substituting $V(x) = 0$ into the equation gives:

$$-\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2} = E\psi(x)$$

To convert this into standard differential equation form, we multiply both sides by $-\frac{2m}{\hbar^2}$:

$$\frac{d^2\psi(x)}{dx^2} = -\frac{2mE}{\hbar^2}\psi(x)$$

Moving all terms to the left-hand side yields:

$$\frac{d^2\psi(x)}{dx^2} + \frac{2mE}{\hbar^2}\psi(x) = 0$$

We define the wave number parameter k through the relation:

$$k^2 = \frac{2mE}{\hbar^2}$$

Substituting k^2 into the differential equation gives the standard harmonic differential equation:

$$\frac{d^2\psi(x)}{dx^2} + k^2\psi(x) = 0$$

Application of Boundary Conditions

The general solution to this second-order linear differential equation is a linear combination of sine and cosine functions:

$$\psi(x) = A\sin(kx) + B\cos(kx)$$

where A and B are constants to be determined by applying boundary conditions and normalization.

For the wave function to be continuous everywhere, it must match the external zero wave function at the boundaries $x = 0$ and $x = a$. The boundary conditions are therefore:

$$\psi(0) = 0$$

$$\psi(a) = 0$$

Applying the first boundary condition at $x = 0$:

$$\psi(0) = A\sin(0) + B\cos(0) = 0$$

Since $\sin(0) = 0$ and $\cos(0) = 1$, this simplifies directly to:

$$B = 0$$

Substituting $B = 0$ back into the general solution leaves:

$$\psi(x) = A\sin(kx)$$

Next, we apply the second boundary condition at $x = a$:

$$\psi(a) = A\sin(ka) = 0$$

To avoid the trivial solution $\psi(x) = 0$ everywhere (which would mean no particle exists inside the box), the constant A cannot be zero. Therefore, we require:

$$\sin(ka) = 0$$

The sine function vanishes at integer multiples of π, so ka must satisfy:

$$ka = n\pi$$

where n is an integer. Setting $n = 0$ yields $k = 0$, leading to $\psi(x) = 0$ everywhere, which is physically unacceptable. Negative integers yield wave functions that differ only by an unphysical overall minus sign from positive integers. Thus, n is restricted to positive integers:

$$n = 1, 2, 3, \ldots$$

Calculation of Energy Eigenvalues

From the boundary condition, the allowed wave numbers k_n are quantized and indexed by n:

$$k_n = \frac{n\pi}{a} \quad \text{for } n = 1, 2, 3, \ldots$$

We can express the energy eigenvalues E_n by using the original definition of k:

$$k_n^2 = \frac{2mE_n}{\hbar^2}$$

Substituting $k_n = \frac{n\pi}{a}$ into this expression gives:

$$\left(\frac{n\pi}{a}\right)^2 = \frac{2mE_n}{\hbar^2}$$

Squaring the left-hand side gives:

$$\frac{n^2\pi^2}{a^2} = \frac{2mE_n}{\hbar^2}$$

Solving explicitly for the energy E_n:

$$E_n = \frac{n^2\pi^2\hbar^2}{2ma^2}$$

We can also express the energy eigenvalues in terms of Planck's constant h. Recalling that $\hbar = \frac{h}{2\pi}$, we square both sides to obtain:

$$\hbar^2 = \frac{h^2}{4\pi^2}$$

Substituting this value of $\hbar^2$ into the expression for E_n:

$$E_n = \frac{n^2\pi^2}{2ma^2}\left(\frac{h^2}{4\pi^2}\right)$$

Canceling π^2 from the numerator and denominator yields the energy eigenvalues:

$$E_n = \frac{n^2h^2}{8ma^2} \quad \text{for } n = 1, 2, 3, \ldots$$

Discrete Energy Spectrum and Zero-Point Energy

The expression for E_n demonstrates that the energy of the particle is quantized; it cannot take any continuous value, but only specific discrete values determined by the integer n, called the principal quantum number.

The lowest possible energy level corresponds to $n = 1$, which is termed the ground state energy or zero-point energy, E_1:

$$E_1 = \frac{h^2}{8ma^2}$$

Because n cannot be zero, the zero-point energy is strictly positive ($E_1 > 0$). A quantum particle confined to a finite box cannot be completely at rest. Higher energy eigenvalues (excited states) can be expressed as integer multiples of the zero-point energy:

$$E_n = n^2 E_1$$

For $n = 2$, the energy of the first excited state is $E_2 = 4E_1$; for $n = 3$, the energy of the second excited state is $E_3 = 9E_1$, and so forth.

Normalization of Eigenfunctions

The wave functions associated with the energy eigenvalues E_n are:

$$\psi_n(x) = A \sin\left(\frac{n\pi x}{a}\right)$$

To determine the constant A, we apply the normalization condition, which states that the total probability of finding the particle inside the well must be equal to unity:

$$\int_0^a |\psi_n(x)|^2 dx = 1$$

Substituting $\psi_n(x)$ into the integral equation:

$$\int_0^a \left| A \sin\left(\frac{n\pi x}{a}\right) \right|^2 dx = 1$$

Assuming A to be real, we pull A^2 outside the integration:

$$A^2 \int_0^a \sin^2\left(\frac{n\pi x}{a}\right) dx = 1$$

We use the trigonometric identity $\sin^2(\theta) = \frac{1-\cos(2\theta)}{2}$ to rewrite the integrand:

$$A^2 \int_0^a \frac{1 - \cos\left(\frac{2n\pi x}{a}\right)}{2} dx = 1$$

Factoring out the constant $\frac{1}{2}$:

$$\frac{A^2}{2} \left[\int_0^a dx - \int_0^a \cos\left(\frac{2n\pi x}{a}\right) dx \right] = 1$$

Evaluating the two integrals separately:

$$\int_0^a dx = a$$

$$\int_0^a \cos\left(\frac{2n\pi x}{a}\right) dx = \left[\frac{a}{2n\pi} \sin\left(\frac{2n\pi x}{a}\right) \right]_0^a$$

Evaluating the second integral at the boundaries $x = a$ and $x = 0$:

$$\frac{a}{2n\pi} \left[\sin(2n\pi) - \sin(0) \right] = 0$$

Substituting these integrated values back into the normalization equation gives:

$$\frac{A^2}{2}(a - 0) = 1$$

$$\frac{A^2 a}{2} = 1$$

Solving for A^2:

$$A^2 = \frac{2}{a}$$

Taking the positive square root gives the normalization constant:

$$A = \sqrt{\frac{2}{a}}$$

Therefore, the fully normalized eigenfunctions $\psi_n(x)$ for a particle in a one-dimensional infinite potential well are:

$$\psi_n(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right) \quad \text{for } 0 < x < a$$

Nodes of Eigenfunctions and Probability Density

The probability density $P_n(x)$ of finding the particle at position x inside the well for state n is given by the square of the normalized wave function:

$$P_n(x) = |\psi_n(x)|^2 = \frac{2}{a} \sin^2\left(\frac{n\pi x}{a}\right)$$

A node is a point inside the open region $0 < x < a$ where the wave function $\psi_n(x)$ equals zero, meaning the probability density vanishes at that location.

To find the nodal positions for state n, we set $\psi_n(x) = 0$:

$$\sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right) = 0$$

This condition is met when the argument of the sine function is an integer multiple of π:

$$\frac{n\pi x}{a} = m\pi \quad \text{for } m = 1, 2, \ldots, n-1$$

Solving for x gives the exact locations of the nodes:

$$x = \frac{ma}{n} \quad \text{for } m = 1, 2, \ldots, n-1$$

Thus, the n-th eigenfunction $\psi_n(x)$ possesses exactly $(n-1)$ nodes inside the well:

For the ground state ($n = 1$), there are $1 - 1 = 0$ nodes inside the well. The wave function $\psi_1(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{\pi x}{a}\right)$ forms a single positive loop, and the probability density $P_1(x) = \frac{2}{a} \sin^2\left(\frac{\pi x}{a}\right)$ reaches its maximum value of $\frac{2}{a}$ at the midpoint of the well, $x = \frac{a}{2}$.

For the first excited state ($n = 2$), there is $2 - 1 = 1$ node located precisely at $x = \frac{a}{2}$. The probability density vanishes at the center of the box and reaches equal maximum values at $x = \frac{a}{4}$ and $x = \frac{3a}{4}$.

For the second excited state ($n = 3$), there are $3 - 1 = 2$ nodes located at $x = \frac{a}{3}$ and $x = \frac{2a}{3}$, with three maximum probability regions situated between the nodes.

2. Particle in a Three-Dimensional Box and Degeneracy

The concept of a particle trapped inside a one-dimensional infinite potential well can be extended to three dimensions. Consider a particle of mass m confined inside a three-dimensional rectangular box of side lengths a, b, and c aligned along the x, y, and z axes respectively. The potential energy function $V(x, y, z)$ describing this system is zero inside the box and infinitely high outside the box:

$$V(x,y,z) = \begin{cases} 0 & \text{for } 0 < x < a,\ 0 < y < b,\ 0 < z < c \\ \infty & \text{otherwise} \end{cases}$$

Since the potential is infinite outside the boundaries, the probability of finding the particle outside the box is zero. Consequently, the wavefunction $\psi(x,y,z)$ must vanish everywhere on and outside the boundaries of the box. Inside the box, the potential energy $V(x,y,z) = 0$, and the time-independent Schrödinger equation takes the form:

$$-\frac{\hbar^2}{2m}\left(\frac{\partial^2\psi}{\partial x^2} + \frac{\partial^2\psi}{\partial y^2} + \frac{\partial^2\psi}{\partial z^2}\right) = E\psi(x,y,z)$$

To clear the fraction, we multiply both sides of the equation by $-\frac{2m}{\hbar^2}$:

$$\frac{\partial^2\psi}{\partial x^2} + \frac{\partial^2\psi}{\partial y^2} + \frac{\partial^2\psi}{\partial z^2} = -\frac{2mE}{\hbar^2}\psi(x,y,z)$$

Rearranging all terms to the left-hand side yields:

$$\frac{\partial^2\psi}{\partial x^2} + \frac{\partial^2\psi}{\partial y^2} + \frac{\partial^2\psi}{\partial z^2} + \frac{2mE}{\hbar^2}\psi(x,y,z) = 0$$

Separation of Variables

To solve this partial differential equation, we apply the method of separation of variables. We assume that the total wavefunction $\psi(x,y,z)$ can be written as a product of three independent single-variable functions:

$$\psi(x,y,z) = X(x)Y(y)Z(z)$$

We substitute this product form of $\psi(x,y,z)$ back into the three-dimensional Schrödinger equation:

$$\frac{\partial^2}{\partial x^2}[X(x)Y(y)Z(z)] + \frac{\partial^2}{\partial y^2}[X(x)Y(y)Z(z)] + \frac{\partial^2}{\partial z^2}[X(x)Y(y)Z(z)] + \frac{2mE}{\hbar^2}X(x)Y(y)Z(z) = 0$$

Carrying out the partial differentiations gives:

$$Y(y)Z(z)\frac{d^2X(x)}{dx^2} + X(x)Z(z)\frac{d^2Y(y)}{dy^2} + X(x)Y(y)\frac{d^2Z(z)}{dz^2} + \frac{2mE}{\hbar^2}X(x)Y(y)Z(z) = 0$$

To separate the variables, we divide the entire equation by the product function $X(x)Y(y)Z(z)$:

$$\frac{1}{X(x)}\frac{d^2X(x)}{dx^2} + \frac{1}{Y(y)}\frac{d^2Y(y)}{dy^2} + \frac{1}{Z(z)}\frac{d^2Z(z)}{dz^2} + \frac{2mE}{\hbar^2} = 0$$

In this equation, each of the first three terms depends solely on a single independent variable (x, y, or z respectively), while the fourth term is a constant. For the sum of these independent terms to equal zero for all arbitrary values of x, y, and z, each individual ratio term must independently equal a constant. We set these separation constants to $-k_x^2$, $-k_y^2$, and $-k_z^2$:

$$\frac{1}{X(x)}\frac{d^2X(x)}{dx^2} = -k_x^2$$

$$\frac{1}{Y(y)}\frac{d^2Y(y)}{dy^2} = -k_y^2$$

$$\frac{1}{Z(z)}\frac{d^2Z(z)}{dz^2} = -k_z^2$$

Substituting these constant definitions into the separated equation gives the condition linking the wave numbers to the total energy:

$$-k_x^2 - k_y^2 - k_z^2 + \frac{2mE}{\hbar^2} = 0$$

Rearranging this relation expresses the total energy E in terms of the individual components:

$$E = \frac{\hbar^2}{2m}\left(k_x^2 + k_y^2 + k_z^2\right)$$

This demonstrates that the total energy E is the sum of three independent energy components: $E = E_x + E_y + E_z$, where $E_x = \frac{\hbar^2 k_x^2}{2m}$, $E_y = \frac{\hbar^2 k_y^2}{2m}$, and $E_z = \frac{\hbar^2 k_z^2}{2m}$.

Solution of Spatial Differential Equations and Boundary Conditions

Each separated ordinary differential equation represents a standard one-dimensional simple harmonic differential equation. Considering the x-dependent equation first:

$$\frac{d^2X(x)}{dx^2} + k_x^2 X(x) = 0$$

The general solution is:

$$X(x) = A_x \sin(k_x x) + B_x \cos(k_x x)$$

We apply the boundary condition that the wavefunction must vanish at the hard wall $x = 0$:

$$X(0) = A_x \sin(0) + B_x \cos(0) = 0$$

Since $\sin(0) = 0$ and $\cos(0) = 1$, this condition directly requires:

$$B_x = 0$$

Substituting $B_x = 0$ simplifies the solution for $X(x)$ to:

$$X(x) = A_x \sin(k_x x)$$

Next, we apply the boundary condition at the opposite wall $x = a$:

$$X(a) = A_x \sin(k_x a) = 0$$

For a non-trivial physical solution ($A_x \neq 0$), the sine term must evaluate to zero:

$$\sin(k_x a) = 0$$

This requires the argument $k_x a$ to be an integer multiple of π:

$$k_x a = n_x \pi \quad \text{where } n_x = 1, 2, 3, \ldots$$

Solving for k_x yields:

$$k_x = \frac{n_x \pi}{a}$$

Following identical mathematical steps and boundary conditions for $Y(y)$ at $y = 0$ and $y = b$, as well as $Z(z)$ at $z = 0$ and $z = c$, yields the corresponding wave numbers:

$$k_y = \frac{n_y \pi}{b} \quad \text{where } n_y = 1, 2, 3, \dots$$

$$k_z = \frac{n_z \pi}{c} \quad \text{where } n_z = 1, 2, 3, \dots$$

Here, n_x, n_y, and n_z are positive integers known as quantum numbers.

Energy Eigenvalues of a Rectangular Box

Substituting the expressions for k_x, k_y, and k_z back into the total energy equation $E = \frac{\hbar^2}{2m}(k_x^2 + k_y^2 + k_z^2)$ gives:

$$E = \frac{\hbar^2}{2m} \left[\left(\frac{n_x \pi}{a}\right)^2 + \left(\frac{n_y \pi}{b}\right)^2 + \left(\frac{n_z \pi}{c}\right)^2 \right]$$

Using the identity $\hbar = \frac{h}{2\pi}$, we square both sides to get $\hbar^2 = \frac{h^2}{4\pi^2}$. Substituting this into the energy expression gives:

$$E = \frac{1}{2m} \left(\frac{h^2}{4\pi^2}\right) \left(\frac{n_x^2 \pi^2}{a^2} + \frac{n_y^2 \pi^2}{b^2} + \frac{n_z^2 \pi^2}{c^2} \right)$$

Factoring out the common term π^2 from the brackets gives:

$$E = \frac{h^2 \pi^2}{8m\pi^2} \left(\frac{n_x^2}{a^2} + \frac{n_y^2}{b^2} + \frac{n_z^2}{c^2} \right)$$

Cancelling π^2 from both the numerator and denominator yields the energy eigenvalues for a particle in a rectangular three-dimensional box:

$$E_{n_x, n_y, n_z} = \frac{h^2}{8m} \left(\frac{n_x^2}{a^2} + \frac{n_y^2}{b^2} + \frac{n_z^2}{c^2} \right)$$

Normalized Eigenfunctions

The total unnormalized eigenfunction $\psi(x, y, z)$ is the product of the three individual spatial solutions:

$$\psi(x, y, z) = A_x A_y A_z \sin\left(\frac{n_x \pi x}{a}\right) \sin\left(\frac{n_y \pi y}{b}\right) \sin\left(\frac{n_z \pi z}{c}\right)$$

Let $A = A_x A_y A_z$ be the overall normalization constant. The normalization requirement over the volume of the box is:

$$\int_0^a \int_0^b \int_0^c |\psi(x, y, z)|^2 \, dx \, dy \, dz = 1$$

Substituting the product wavefunction into this integral allows it to factorize into three independent one-dimensional integrals:

$$A^2 \left[\int_0^a \sin^2\left(\frac{n_x \pi x}{a}\right) dx\right] \left[\int_0^b \sin^2\left(\frac{n_y \pi y}{b}\right) dy\right] \left[\int_0^c \sin^2\left(\frac{n_z \pi z}{c}\right) dz\right] = 1$$

Using the trigonometric identity $\sin^2\theta = \frac{1-\cos(2\theta)}{2}$, the first integral evaluates to:

$$\int_0^a \sin^2\left(\frac{n_x \pi x}{a}\right) dx = \int_0^a \frac{1 - \cos\left(\frac{2n_x \pi x}{a}\right)}{2} dx = \left[\frac{x}{2} - \frac{a}{4n_x \pi} \sin\left(\frac{2n_x \pi x}{a}\right)\right]_0^a = \frac{a}{2}$$

Similarly, the second and third integrals evaluate to:

$$\int_0^b \sin^2\left(\frac{n_y \pi y}{b}\right) dy = \frac{b}{2}$$

$$\int_0^c \sin^2\left(\frac{n_z \pi z}{c}\right) dz = \frac{c}{2}$$

Substituting these individual integral results into the normalization condition gives:

$$A^2 \left(\frac{a}{2}\right) \left(\frac{b}{2}\right) \left(\frac{c}{2}\right) = 1$$

Simplifying the algebraic product:

$$A^2 \frac{abc}{8} = 1$$

Solving for A yields:

$$A = \sqrt{\frac{8}{abc}}$$

Therefore, the fully normalized eigenfunctions for a particle in a three-dimensional rectangular box are given by:

$$\psi_{n_x,n_y,n_z}(x,y,z) = \sqrt{\frac{8}{abc}} \sin\left(\frac{n_x \pi x}{a}\right) \sin\left(\frac{n_y \pi y}{b}\right) \sin\left(\frac{n_z \pi z}{c}\right)$$

Special Case: Particle in a Cubical Box

For a cubical box, all three sides are equal in length, so we set $a = b = c$. Substituting $b = a$ and $c = a$ into the general energy eigenvalue equation gives:

$$E = \frac{h^2}{8m} \left(\frac{n_x^2}{a^2} + \frac{n_y^2}{a^2} + \frac{n_z^2}{a^2}\right)$$

Factoring out the constant denominator a^2 gives the energy eigenvalues for a cubical box of edge length a:

$$E_{n_x,n_y,n_z} = \frac{h^2}{8ma^2} \left(n_x^2 + n_y^2 + n_z^2\right)$$

Similarly, substituting $b = a$ and $c = a$ into the normalized eigenfunction formula yields:

$$\psi_{n_x,n_y,n_z}(x,y,z) = \sqrt{\frac{8}{a^3}} \sin\left(\frac{n_x \pi x}{a}\right) \sin\left(\frac{n_y \pi y}{a}\right) \sin\left(\frac{n_z \pi z}{a}\right)$$

Concept of Degeneracy

In quantum mechanics, a state is said to be degenerate when two or more distinct, linearly independent eigenfunctions share the exact same energy eigenvalue. The number of independent eigenfunctions corresponding to a single energy level is defined as the degree of degeneracy (or degeneracy factor, g) of that energy level. If only one eigenfunction corresponds to a particular energy value, the state is called non-degenerate ($g = 1$).

In a cubical box, the high degree of spatial symmetry ($a = b = c$) causes energy eigenvalues to depend only on the sum of the squares of the quantum numbers, $(n_x^2 + n_y^2 + n_z^2)$. Consequently, different permutations of the set of quantum numbers (n_x, n_y, n_z) result in distinct spatial wavefunctions $\psi_{n_x,n_y,n_z}(x, y, z)$, but yield the exact same total energy value E.

Let $E_0 = \frac{h^2}{8ma^2}$ be the baseline ground-state unit of energy for a cubical box. The energy of any state (n_x, n_y, n_z) can be expressed as $E = (n_x^2 + n_y^2 + n_z^2)E_0$. The table below summarizes the quantum states, their corresponding energy values, and their degrees of degeneracy for the lowest energy levels of a particle in a cubical box:

Quantum Numbers (n_x, n_y, n_z)	$n_x^2 + n_y^2 + n_z^2$	**Energy (** E**)**	**Distinct Eigenfunctions**	**Degree of Degeneracy (**g**)**
$(1, 1, 1)$	$1^2 + 1^2 + 1^2 = 3$	$3E_0$	$\psi_{1,1,1}$	1 (Non-degenerate)
$(1, 1, 2), (1, 2, 1), (2, 1, 1)$	$1^2 + 1^2 + 2^2 = 6$	$6E_0$	$\psi_{1,1,2}, \psi_{1,2,1}, \psi_{2,1,1}$	3 (Three-fold degenerate)
$(1, 2, 2), (2, 1, 2), (2, 2, 1)$	$1^2 + 2^2 + 2^2 = 9$	$9E_0$	$\psi_{1,2,2}, \psi_{2,1,2}, \psi_{2,2,1}$	3 (Three-fold degenerate)
$(1, 1, 3), (1, 3, 1), (3, 1, 1)$	$1^2 + 1^2 + 3^2 = 11$	$11E_0$	$\psi_{1,1,3}, \psi_{1,3,1}, \psi_{3,1,1}$	3 (Three-fold degenerate)
$(2, 2, 2)$	$2^2 + 2^2 + 2^2 = 12$	$12E_0$	$\psi_{2,2,2}$	1 (Non-degenerate)

Calculation of Degeneracy for Low-Lying Energy Levels

To demonstrate the calculation of the degree of degeneracy, consider specific low-lying energy levels of a particle in a cubical box:

For the ground state, the quantum numbers are $n_x = 1, n_y = 1, n_z = 1$. The sum of squares is:

$$n_x^2 + n_y^2 + n_z^2 = 1^2 + 1^2 + 1^2 = 3$$

The energy is $E_{1,1,1} = 3E_0$. Because all three quantum numbers are identical, no alternative permutations exist. There is only one distinct wavefunction $\psi_{1,1,1}$, so the ground state has a degree of degeneracy $g = 1$ (non-degenerate).

For the first excited state, the set of quantum numbers consists of two 1s and one 2. The sum of squares is:

$$n_x^2 + n_y^2 + n_z^2 = 1^2 + 1^2 + 2^2 = 6$$

The energy is $E = 6E_0$. The three distinct permutations of these quantum numbers are $(1, 1, 2)$, $(1, 2, 1)$, and $(2, 1, 1)$. Each set gives a different wavefunction:

$$\psi_{1,1,2} = \sqrt{\frac{8}{a^3}} \sin\left(\frac{\pi x}{a}\right) \sin\left(\frac{\pi y}{a}\right) \sin\left(\frac{2\pi z}{a}\right)$$

$$\psi_{1,2,1} = \sqrt{\frac{8}{a^3}} \sin\left(\frac{\pi x}{a}\right) \sin\left(\frac{2\pi y}{a}\right) \sin\left(\frac{\pi z}{a}\right)$$

$$\psi_{2,1,1} = \sqrt{\frac{8}{a^3}} \sin\left(\frac{2\pi x}{a}\right) \sin\left(\frac{\pi y}{a}\right) \sin\left(\frac{\pi z}{a}\right)$$

Since three independent wavefunctions produce the identical energy eigenvalue $6E_0$, the degree of degeneracy for this energy level is $g = 3$ (three-fold degenerate).

For the state with quantum numbers composed of two 2s and one 1, the sum of squares is:

$$n_x^2 + n_y^2 + n_z^2 = 1^2 + 2^2 + 2^2 = 9$$

The energy is $E = 9E_0$. Permutations of $(1, 2, 2)$ yield three distinct combinations: $(1, 2, 2)$, $(2, 1, 2)$, and $(2, 2, 1)$. Thus, there are three linearly independent eigenfunctions ($\psi_{1,2,2}, \psi_{2,1,2}, \psi_{2,2,1}$) possessing the exact same energy, giving a degree of degeneracy $g = 3$.

For the state with quantum numbers $(2, 2, 2)$, the sum of squares is:

$$n_x^2 + n_y^2 + n_z^2 = 2^2 + 2^2 + 2^2 = 12$$

The energy is $E = 12E_0$. Since $n_x = n_y = n_z = 2$, permutating the quantum numbers yields no new combinations. Only one single wavefunction $\psi_{2,2,2}$ corresponds to $E = 12E_0$, making this energy level non-degenerate ($g = 1$).

3. Step Potential: Transmission and Reflection Coefficients

Consider a particle of mass m and total energy E moving in one dimension under the influence of a step potential $V(x)$. The potential profile is defined mathematically as:

$$V(x) = \begin{cases} 0 & \text{for } x < 0 \quad \text{(Region I)} \\ V_0 & \text{for } x > 0 \quad \text{(Region II)} \end{cases}$$

where V_0 is a positive constant representing the height of the potential step located at the boundary $x = 0$. A beam of particles is incident on the potential barrier from the left ($x < 0$). We analyze the system for two distinct energy regimes: $E > V_0$ and $E < V_0$.

Case 1: Incident Energy Greater than Step Height ($E > V_0$)

When the total energy E of the incident particle exceeds the potential height V_0, the particle has sufficient energy to move in both regions according to classical mechanics.

In Region I ($x < 0$), the potential is $V(x) = 0$. The time-independent Schrödinger equation is given by:

$$-\frac{\hbar^2}{2m}\frac{d^2\psi_I(x)}{dx^2} = E\psi_I(x)$$

Rearranging this differential equation into standard form yields:

$$\frac{d^2\psi_I(x)}{dx^2} + \frac{2mE}{\hbar^2}\psi_I(x) = 0$$

We define the wave vector k_1 in Region I as:

$$k_1 = \frac{\sqrt{2mE}}{\hbar}$$

Substituting k_1 into the differential equation yields:

$$\frac{d^2\psi_I(x)}{dx^2} + k_1^2\psi_I(x) = 0$$

The general solution to this second-order differential equation is a linear combination of complex exponentials:

$$\psi_I(x) = Ae^{ik_1x} + Be^{-ik_1x}$$

In this expression, Ae^{ik_1x} represents the incident wave propagating to the right (in the $+x$ direction) with amplitude A, and Be^{-ik_1x} represents the reflected wave propagating to the left (in the $-x$ direction) with amplitude B.

In Region II ($x > 0$), the potential is $V(x) = V_0$. The time-independent Schrödinger equation is given by:

$$-\frac{\hbar^2}{2m}\frac{d^2\psi_{II}(x)}{dx^2} + V_0\psi_{II}(x) = E\psi_{II}(x)$$

Rearranging terms to isolate the second derivative yields:

$$\frac{d^2\psi_{II}(x)}{dx^2} + \frac{2m(E-V_0)}{\hbar^2}\psi_{II}(x) = 0$$

We define the wave vector k_2 in Region II as:

$$k_2 = \frac{\sqrt{2m(E-V_0)}}{\hbar}$$

Substituting k_2 into the differential equation gives:

$$\frac{d^2\psi_{II}(x)}{dx^2} + k_2^2\psi_{II}(x) = 0$$

The general solution in Region II is given by:

$$\psi_{II}(x) = Ce^{ik_2x} + De^{-ik_2x}$$

Since particles are incident only from the left ($x < 0$), there is no mechanism to produce a wave traveling to the left in Region II. Consequently, the amplitude of the left-traveling wave in Region II must be zero, so $D = 0$. The wave function in Region II simplifies to:

$$\psi_{II}(x) = Ce^{ik_2x}$$

Here, Ce^{ik_2x} represents the transmitted wave propagating to the right with amplitude C.

To determine the relations between the coefficients A, B, and C, we apply boundary conditions at $x = 0$. Physical wave functions must be continuous and have continuous first derivatives across any finite potential boundary.

First, applying the continuity of the wave function at $x = 0$, $\psi_I(0) = \psi_{II}(0)$, gives:

$$Ae^{ik_1(0)} + Be^{-ik_1(0)} = Ce^{ik_2(0)}$$

Evaluating the exponential factors at zero yields:

$$A + B = C$$

Second, taking the first derivative of the wave functions in both regions yields:

$$\frac{d\psi_I}{dx} = ik_1Ae^{ik_1x} - ik_1Be^{-ik_1x}$$

$$\frac{d\psi_{II}}{dx} = ik_2Ce^{ik_2x}$$

Applying the continuity of the derivative at $x = 0$, $\left.\frac{d\psi_I}{dx}\right|_{x=0} = \left.\frac{d\psi_{II}}{dx}\right|_{x=0}$, gives:

$$ik_1A - ik_1B = ik_2C$$

Dividing both sides of this equation by ik_1 yields:

$$A - B = \frac{k_2}{k_1}C$$

To find the transmitted amplitude C in terms of the incident amplitude A, we add the continuity equation $A + B = C$ to the derivative equation $A - B = \frac{k_2}{k_1}C$:

$$(A + B) + (A - B) = C + \frac{k_2}{k_1}C$$

Combining like terms gives:

$$2A = \left(1 + \frac{k_2}{k_1}\right)C$$

Finding a common denominator inside the parentheses yields:

$$2A = \left(\frac{k_1 + k_2}{k_1}\right)C$$

Solving for the ratio C/A gives the relative transmission amplitude:

$$\frac{C}{A} = \frac{2k_1}{k_1 + k_2}$$

To find the reflected amplitude B in terms of the incident amplitude A, we subtract the derivative equation from the continuity equation:

$$(A + B) - (A - B) = C - \frac{k_2}{k_1}C$$

Simplifying the left-hand side yields:

$$2B = \left(1 - \frac{k_2}{k_1}\right)C$$

Writing the terms inside the parentheses with a common denominator gives:

$$2B = \left(\frac{k_1 - k_2}{k_1}\right) C$$

Substituting $C = \frac{2k_1 A}{k_1 + k_2}$ into this expression gives:

$$2B = \left(\frac{k_1 - k_2}{k_1}\right) \left(\frac{2k_1 A}{k_1 + k_2}\right)$$

Canceling $2k_1$ from both sides yields:

$$B = \frac{k_1 - k_2}{k_1 + k_2} A$$

Dividing both sides by A gives the relative reflection amplitude:

$$\frac{B}{A} = \frac{k_1 - k_2}{k_1 + k_2}$$

To determine the reflection and transmission coefficients, we calculate the probability current density J, defined in one dimension as:

$$J = \frac{\hbar}{2mi}\left(\psi^* \frac{d\psi}{dx} - \psi \frac{d\psi^*}{dx}\right)$$

Evaluating J for the incident wave $\psi_{inc} = Ae^{ik_1 x}$ gives:

$$J_{inc} = \frac{\hbar}{2mi}\left(A^* e^{-ik_1 x}(ik_1 A e^{ik_1 x}) - Ae^{ik_1 x}(-ik_1 A^* e^{-ik_1 x})\right)$$

Simplifying the products yields:

$$J_{inc} = \frac{\hbar}{2mi}\left(ik_1|A|^2 + ik_1|A|^2\right) = \frac{\hbar k_1}{m}|A|^2$$

Evaluating J for the reflected wave $\psi_{ref} = Be^{-ik_1 x}$ gives (the minus sign shows it flows to the left):

$$J_{ref} = -\frac{\hbar k_1}{m}|B|^2$$

Evaluating J for the transmitted wave $\psi_{trans} = Ce^{ik_2 x}$ gives:

$$J_{trans} = \frac{\hbar k_2}{m}|C|^2$$

The reflection coefficient R is defined as the ratio of the magnitude of the reflected probability current density to the incident probability current density:

$$R = \frac{|J_{ref}|}{J_{inc}} = \frac{\frac{\hbar k_1}{m}|B|^2}{\frac{\hbar k_1}{m}|A|^2}$$

Canceling the factor $\frac{\hbar k_1}{m}$ yields:

$$R = \left| \frac{B}{A} \right|^2$$

Substituting the ratio $\frac{B}{A} = \frac{k_1 - k_2}{k_1 + k_2}$ into the definition of R gives:

$$R = \left(\frac{k_1 - k_2}{k_1 + k_2} \right)^2$$

The transmission coefficient T is defined as the ratio of the transmitted probability current density to the incident probability current density:

$$T = \frac{J_{trans}}{J_{inc}} = \frac{\frac{\hbar k_2}{m} |C|^2}{\frac{\hbar k_1}{m} |A|^2}$$

Canceling the common factor $\frac{\hbar}{m}$ yields:

$$T = \frac{k_2}{k_1} \left| \frac{C}{A} \right|^2$$

Substituting the ratio $\frac{C}{A} = \frac{2k_1}{k_1 + k_2}$ into the definition of T gives:

$$T = \frac{k_2}{k_1} \left(\frac{2k_1}{k_1 + k_2} \right)^2$$

Squaring the fraction inside the parentheses yields:

$$T = \frac{k_2}{k_1} \frac{4k_1^2}{(k_1 + k_2)^2}$$

Canceling k_1 in the numerator and denominator gives:

$$T = \frac{4k_1 k_2}{(k_1 + k_2)^2}$$

To verify probability conservation, we sum the reflection coefficient R and the transmission coefficient T:

$$R + T = \left(\frac{k_1 - k_2}{k_1 + k_2} \right)^2 + \frac{4k_1 k_2}{(k_1 + k_2)^2}$$

Combining both terms over the common denominator $(k_1 + k_2)^2$ gives:

$$R + T = \frac{(k_1 - k_2)^2 + 4k_1 k_2}{(k_1 + k_2)^2}$$

Expanding the squared term in the numerator yields:

$$R + T = \frac{k_1^2 - 2k_1 k_2 + k_2^2 + 4k_1 k_2}{(k_1 + k_2)^2}$$

Combining the middle terms $-2k_1 k_2 + 4k_1 k_2 = 2k_1 k_2$ gives:

$$R + T = \frac{k_1^2 + 2k_1 k_2 + k_2^2}{(k_1 + k_2)^2}$$

Factoring the numerator into a perfect square yields:

$$R + T = \frac{(k_1 + k_2)^2}{(k_1 + k_2)^2}$$

Simplifying this expression gives:

$$R + T = 1$$

This demonstrates that total probability current is strictly conserved across the step potential boundary.

Case 2: Incident Energy Less than Step Height ($E < V_0$)

When the incident energy E is less than the potential step height V_0, Region II ($x > 0$) is classically forbidden, as a classical particle would have a negative kinetic energy.

In Region I ($x < 0$), where $V(x) = 0$, the time-independent Schrödinger equation remains unchanged from Case 1:

$$\frac{d^2\psi_I(x)}{dx^2} + k_1^2\psi_I(x) = 0$$

where the wave vector k_1 is:

$$k_1 = \frac{\sqrt{2mE}}{\hbar}$$

The wave function in Region I is:

$$\psi_I(x) = Ae^{ik_1x} + Be^{-ik_1x}$$

In Region II ($x > 0$), where $V(x) = V_0 > E$, the Schrödinger equation is:

$$-\frac{\hbar^2}{2m}\frac{d^2\psi_{II}(x)}{dx^2} + V_0\psi_{II}(x) = E\psi_{II}(x)$$

Rearranging terms yields:

$$\frac{d^2\psi_{II}(x)}{dx^2} - \frac{2m(V_0 - E)}{\hbar^2}\psi_{II}(x) = 0$$

Because $V_0 > E$, the quantity $V_0 - E$ is real and positive. We define a real attenuation constant α as:

$$\alpha = \frac{\sqrt{2m(V_0 - E)}}{\hbar}$$

Substituting α into the differential equation gives:

$$\frac{d^2\psi_{II}(x)}{dx^2} - \alpha^2\psi_{II}(x) = 0$$

The general solution to this differential equation is:

$$\psi_{II}(x) = Ce^{-\alpha x} + De^{\alpha x}$$

As $x \to \infty$, the term $e^{\alpha x}$ diverges to infinity, which violates the normalization condition that $\psi_{II}(x)$ must remain finite everywhere. Thus, we set $D = 0$, leaving:

$$\psi_{II}(x) = Ce^{-\alpha x}$$

This result demonstrates that the wave function penetrates into the classically forbidden region ($x > 0$), decaying exponentially with distance.

To evaluate the coefficients, we apply the boundary conditions at $x = 0$. First, continuity of the wave function $\psi_I(0) = \psi_{II}(0)$ gives:

$$A + B = C$$

Second, taking the first derivative of the wave functions:

$$\frac{d\psi_I}{dx} = ik_1Ae^{ik_1x} - ik_1Be^{-ik_1x}$$

$$\frac{d\psi_{II}}{dx} = -\alpha Ce^{-\alpha x}$$

Applying continuity of the derivative at $x = 0$ yields:

$$ik_1A - ik_1B = -\alpha C$$

Dividing both sides by ik_1 gives:

$$A - B = -\frac{\alpha}{ik_1}C$$

Using the identity $1/i = -i$, this simplifies to:

$$A - B = \frac{i\alpha}{k_1}C$$

Dividing the derivative condition $(A - B) = \frac{i\alpha}{k_1}C$ by the wave function continuity condition $(A + B) = C$ cancels C:

$$\frac{A - B}{A + B} = \frac{i\alpha}{k_1}$$

Cross-multiplying gives:

$$k_1(A - B) = i\alpha(A + B)$$

Expanding both sides yields:

$$k_1A - k_1B = i\alpha A + i\alpha B$$

Grouping terms containing A on the left and terms containing B on the right gives:

$$(k_1 - i\alpha)A = (k_1 + i\alpha)B$$

Solving for the ratio B/A yields:

$$\frac{B}{A} = \frac{k_1 - i\alpha}{k_1 + i\alpha}$$

The reflection coefficient R is calculated as the modulus squared of the amplitude ratio B/A:

$$R = \left|\frac{B}{A}\right|^2 = \left(\frac{B}{A}\right)\left(\frac{B}{A}\right)^*$$

Taking the complex conjugate of B/A gives:

$$\left(\frac{B}{A}\right)^* = \frac{k_1 + i\alpha}{k_1 - i\alpha}$$

Multiplying the ratio by its complex conjugate gives:

$$R = \left(\frac{k_1 - i\alpha}{k_1 + i\alpha}\right)\left(\frac{k_1 + i\alpha}{k_1 - i\alpha}\right)$$

Multiplying the numerators and denominators gives:

$$R = \frac{k_1^2 + \alpha^2}{k_1^2 + \alpha^2}$$

Simplifying this expression yields:

$$R = 1$$

Since $R = 1$, the transmission coefficient T must be:

$$T = 1 - R = 0$$

Although $T = 0$ (meaning no net probability flux is transported to $+\infty$), the particle has a non-zero probability density $|\psi_{II}(x)|^2 = |C|^2 e^{-2\alpha x}$ of being found inside the region $x > 0$.

Physical Discussion

The quantum mechanical step potential reveals fundamental differences from classical mechanics:

1. **Partial Reflection for $E > V_0$**: Classically, a particle with energy $E > V_0$ passing over a potential step experiences a decrease in velocity but is transmitted into Region II with 100% certainty ($T = 1, R = 0$). In quantum mechanics, because matter behaves as a wave, a sudden change in the potential $V(x)$ corresponds to a change in the refractive index for matter waves (altering the wavenumber from k_1 to k_2). This sharp discontinuity causes partial reflection ($R > 0$) at the boundary $x = 0$, even though the particle possesses more energy than the step height.

1. **Penetration for $E < V_0$**: Classically, a particle with energy $E < V_0$ cannot cross $x = 0$ at all; it is reflected instantaneously with zero probability of entering Region II. Quantum mechanically, while all incident particles are ultimately reflected ($R = 1, T = 0$), the wave function does not drop abruptly to zero at the boundary. Instead, it penetrates into the classically forbidden region ($x > 0$), decaying exponentially over a characteristic depth $\delta = 1/\alpha$.

4. Finite Rectangular Potential Barrier and Quantum Tunnelling

Consider a particle of mass m moving in one dimension under the influence of a finite rectangular potential barrier. The potential energy function $V(x)$ is defined as

$$V(x) = \begin{cases} 0, & x < 0 \quad \text{(Region I)} \\ V_0, & 0 < x < L \quad \text{(Region II)} \\ 0, & x > L \quad \text{(Region III)} \end{cases}$$

where $V_0 > 0$ represents the height of the potential barrier and L represents its width.

We consider the case where the total energy E of the particle is less than the barrier height, that is, $E < V_0$. Classically, a particle approaching the barrier from the left with energy $E < V_0$ cannot enter Region II because its kinetic energy inside the barrier would be $E - V_0 < 0$, which is impossible. Therefore, classical mechanics predicts total reflection at $x = 0$. In quantum mechanics, however, the wave function can penetrate the barrier region and remain non-zero at $x = L$, leading to a finite probability of the particle appearing in Region III.

Wave Functions in the Three Regions

The time-independent Schrödinger equation for a particle of mass m with energy E in a potential $V(x)$ is given by

$$-\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2} + V(x)\psi(x) = E\psi(x)$$

We divide the space into three distinct regions to solve this differential equation.

In Region I ($x < 0$), the potential is $V(x) = 0$. Substituting $V(x) = 0$ into the Schrödinger equation yields

$$-\frac{\hbar^2}{2m}\frac{d^2\psi_I(x)}{dx^2} = E\psi_I(x)$$

Rearranging terms gives

$$\frac{d^2\psi_I(x)}{dx^2} + \frac{2mE}{\hbar^2}\psi_I(x) = 0$$

We define the wave vector k_1 in Region I as

$$k_1 = \frac{\sqrt{2mE}}{\hbar}$$

Substituting k_1 into the differential equation gives

$$\frac{d^2\psi_I(x)}{dx^2} + k_1^2\psi_I(x) = 0$$

The general solution of this equation is a combination of incident and reflected plane waves:

$$\psi_I(x) = Ae^{ik_1x} + Be^{-ik_1x}$$

where Ae^{ik_1x} represents the incident wave propagating in the positive x-direction and Be^{-ik_1x} represents the reflected wave propagating in the negative x-direction.

In Region II ($0 < x < L$), the potential is constant at $V(x) = V_0$. Substituting $V(x) = V_0$ into the Schrödinger equation yields

$$-\frac{\hbar^2}{2m}\frac{d^2\psi_{II}(x)}{dx^2} + V_0\psi_{II}(x) = E\psi_{II}(x)$$

Rearranging the terms gives

$$\frac{d^2\psi_{II}(x)}{dx^2} - \frac{2m(V_0 - E)}{\hbar^2}\psi_{II}(x) = 0$$

Since $E < V_0$, the quantity $(V_0 - E)$ is positive. We define the attenuation parameter κ as

$$\kappa = \frac{\sqrt{2m(V_0 - E)}}{\hbar}$$

Substituting κ into the differential equation gives

$$\frac{d^2\psi_{II}(x)}{dx^2} - \kappa^2\psi_{II}(x) = 0$$

The general solution of this equation inside the barrier is given by real exponential functions:

$$\psi_{II}(x) = Ce^{\kappa x} + De^{-\kappa x}$$

where C and D are constants to be determined by boundary conditions.

In Region III ($x > L$), the potential returns to $V(x) = 0$. The Schrödinger equation is identical in form to that in Region I:

$$\frac{d^2\psi_{III}(x)}{dx^2} + k_1^2\psi_{III}(x) = 0$$

The general solution is

$$\psi_{III}(x) = Fe^{ik_1 x} + Ge^{-ik_1 x}$$

Because the particle is incident from the left, there is no physical mechanism to reflect waves back to the left in Region III. Therefore, we set $G = 0$, giving the transmitted wave function:

$$\psi_{III}(x) = Fe^{ik_1 x}$$

where $Fe^{ik_1 x}$ represents the transmitted wave moving to the right.

Application of Boundary Conditions

To ensure physical acceptability, the wave function $\psi(x)$ and its first derivative $\frac{d\psi(x)}{dx}$ must be continuous across the potential boundaries at $x = 0$ and $x = L$.

First, applying the continuity of the wave function at $x = 0$:

$$\psi_I(0) = \psi_{II}(0)$$

Substituting the forms of $\psi_I(x)$ and $\psi_{II}(x)$ evaluated at $x = 0$ gives

$$A + B = C + D$$

Second, applying the continuity of the first derivative of the wave function at $x = 0$:

$$\left.\frac{d\psi_I}{dx}\right|_{x=0} = \left.\frac{d\psi_{II}}{dx}\right|_{x=0}$$

Differentiating $\psi_I(x)$ and $\psi_{II}(x)$ with respect to x and evaluating at $x = 0$ yields

$$ik_1A - ik_1B = \kappa C - \kappa D$$

Dividing both sides by ik_1 gives

$$A - B = \frac{\kappa}{ik_1}(C - D)$$

Using the identity $\frac{1}{i} = -i$, this expression becomes

$$A - B = -i\frac{\kappa}{k_1}(C - D)$$

Third, applying the continuity of the wave function at $x = L$:

$$\psi_{II}(L) = \psi_{III}(L)$$

Substituting the forms of $\psi_{II}(x)$ and $\psi_{III}(x)$ evaluated at $x = L$ gives

$$Ce^{\kappa L} + De^{-\kappa L} = Fe^{ik_1L}$$

Fourth, applying the continuity of the first derivative of the wave function at $x = L$:

$$\left.\frac{d\psi_{II}}{dx}\right|_{x=L} = \left.\frac{d\psi_{III}}{dx}\right|_{x=L}$$

Differentiating $\psi_{II}(x)$ and $\psi_{III}(x)$ with respect to x and evaluating at $x = L$ yields

$$\kappa Ce^{\kappa L} - \kappa De^{-\kappa L} = ik_1Fe^{ik_1L}$$

Dividing both sides by κ gives

$$Ce^{\kappa L} - De^{-\kappa L} = i\frac{k_1}{\kappa}Fe^{ik_1L}$$

Derivation of the Transmission Coefficient

The transmission coefficient T is defined as the ratio of the transmitted probability current density to the incident probability current density. Because the velocity of the wave is the same in Region I and Region III, T simplifies to the ratio of the square moduli of the amplitudes:

$$T = \frac{|F|^2}{|A|^2} = \left|\frac{F}{A}\right|^2$$

To express A in terms of F, we first solve for C and D in terms of F using the boundary conditions at $x = L$.

Adding the third boundary condition equation to the fourth boundary condition equation gives

$$2Ce^{\kappa L} = \left(1 + i\frac{k_1}{\kappa}\right)Fe^{ik_1L}$$

Multiplying both sides by $\frac{1}{2}e^{-\kappa L}$ isolates C:

$$C = \frac{1}{2}\left(1 + i\frac{k_1}{\kappa}\right)Fe^{(ik_1-\kappa)L}$$

Subtracting the fourth boundary condition equation from the third boundary condition equation gives

$$2De^{-\kappa L} = \left(1 - i\frac{k_1}{\kappa}\right) Fe^{ik_1 L}$$

Multiplying both sides by $\frac{1}{2}e^{\kappa L}$ isolates D:

$$D = \frac{1}{2}\left(1 - i\frac{k_1}{\kappa}\right) Fe^{(ik_1+\kappa)L}$$

Now, adding the first boundary condition equation $A + B = C + D$ to the second boundary condition equation $A - B = -i\frac{\kappa}{k_1}(C - D)$ eliminates B:

$$2A = (C + D) - i\frac{\kappa}{k_1}(C - D)$$

Grouping the terms containing C and D gives

$$2A = \left(1 - i\frac{\kappa}{k_1}\right) C + \left(1 + i\frac{\kappa}{k_1}\right) D$$

Dividing both sides by 2 yields

$$A = \frac{1}{2}\left(1 - i\frac{\kappa}{k_1}\right) C + \frac{1}{2}\left(1 + i\frac{\kappa}{k_1}\right) D$$

Substituting the expressions for C and D in terms of F into the expression for A gives

$$A = \frac{1}{4}\left(1 - i\frac{\kappa}{k_1}\right)\left(1 + i\frac{k_1}{\kappa}\right) Fe^{(ik_1-\kappa)L} + \frac{1}{4}\left(1 + i\frac{\kappa}{k_1}\right)\left(1 - i\frac{k_1}{\kappa}\right) Fe^{(ik_1+\kappa)L}$$

Expanding the product of the factors in the first term yields

$$\left(1 - i\frac{\kappa}{k_1}\right)\left(1 + i\frac{k_1}{\kappa}\right) = 1 + i\frac{k_1}{\kappa} - i\frac{\kappa}{k_1} + 1 = 2 + i\left(\frac{k_1}{\kappa} - \frac{\kappa}{k_1}\right)$$

Expanding the product of the factors in the second term yields

$$\left(1 + i\frac{\kappa}{k_1}\right)\left(1 - i\frac{k_1}{\kappa}\right) = 1 - i\frac{k_1}{\kappa} + i\frac{\kappa}{k_1} + 1 = 2 - i\left(\frac{k_1}{\kappa} - \frac{\kappa}{k_1}\right)$$

Substituting these expanded products back into the equation for A gives

$$A = \frac{Fe^{ik_1 L}}{4}\left[\left(2 + i\left(\frac{k_1}{\kappa} - \frac{\kappa}{k_1}\right)\right)e^{-\kappa L} + \left(2 - i\left(\frac{k_1}{\kappa} - \frac{\kappa}{k_1}\right)\right)e^{\kappa L}\right]$$

Rearranging the terms inside the square bracket according to powers of the exponentials gives

$$A = \frac{Fe^{ik_1 L}}{4}\left[2(e^{\kappa L} + e^{-\kappa L}) - i\left(\frac{k_1}{\kappa} - \frac{\kappa}{k_1}\right)(e^{\kappa L} - e^{-\kappa L})\right]$$

Recalling the definitions of hyperbolic functions:

$$\cosh(\kappa L) = \frac{e^{\kappa L} + e^{-\kappa L}}{2} \implies e^{\kappa L} + e^{-\kappa L} = 2\cosh(\kappa L)$$

$$\sinh(\kappa L) = \frac{e^{\kappa L} - e^{-\kappa L}}{2} \implies e^{\kappa L} - e^{-\kappa L} = 2\sinh(\kappa L)$$

Substituting these hyperbolic identities into the equation for A yields

$$A = \frac{Fe^{ik_1 L}}{4}\left[4\cosh(\kappa L) - 2i\left(\frac{k_1^2 - \kappa^2}{k_1\kappa}\right)\sinh(\kappa L)\right]$$

Dividing the terms inside the bracket by 4 gives

$$A = Fe^{ik_1 L}\left[\cosh(\kappa L) + i\left(\frac{\kappa^2 - k_1^2}{2k_1\kappa}\right)\sinh(\kappa L)\right]$$

To calculate $|A|^2$, we multiply A by its complex conjugate A^*:

$$|A|^2 = |F|^2|e^{ik_1 L}|^2\left[\cosh^2(\kappa L) + \left(\frac{\kappa^2 - k_1^2}{2k_1\kappa}\right)^2\sinh^2(\kappa L)\right]$$

Since $|e^{ik_1 L}|^2 = 1$, the ratio $\left|\frac{A}{F}\right|^2$ becomes

$$\left|\frac{A}{F}\right|^2 = \cosh^2(\kappa L) + \frac{(\kappa^2 - k_1^2)^2}{4k_1^2\kappa^2}\sinh^2(\kappa L)$$

Using the trigonometric/hyperbolic identity $\cosh^2(\kappa L) = 1 + \sinh^2(\kappa L)$, we substitute for $\cosh^2(\kappa L)$:

$$\left|\frac{A}{F}\right|^2 = 1 + \sinh^2(\kappa L) + \frac{(\kappa^2 - k_1^2)^2}{4k_1^2\kappa^2}\sinh^2(\kappa L)$$

Factoring out $\sinh^2(\kappa L)$ yields

$$\left|\frac{A}{F}\right|^2 = 1 + \left[1 + \frac{(\kappa^2 - k_1^2)^2}{4k_1^2\kappa^2}\right]\sinh^2(\kappa L)$$

Combining the terms inside the square bracket over a common denominator gives

$$1 + \frac{(\kappa^2 - k_1^2)^2}{4k_1^2\kappa^2} = \frac{4k_1^2\kappa^2 + \kappa^4 - 2k_1^2\kappa^2 + k_1^4}{4k_1^2\kappa^2} = \frac{\kappa^4 + 2k_1^2\kappa^2 + k_1^4}{4k_1^2\kappa^2} = \frac{(\kappa^2 + k_1^2)^2}{4k_1^2\kappa^2}$$

Now we substitute the definitions $k_1^2 = \frac{2mE}{\hbar^2}$ and $\kappa^2 = \frac{2m(V_0 - E)}{\hbar^2}$ into the numerator $\kappa^2 + k_1^2$:

$$\kappa^2 + k_1^2 = \frac{2m(V_0 - E)}{\hbar^2} + \frac{2mE}{\hbar^2} = \frac{2mV_0}{\hbar^2}$$

Evaluating the entire fraction in terms of energy E and barrier height V_0:

$$\frac{(\kappa^2 + k_1^2)^2}{4k_1^2\kappa^2} = \frac{\left(\frac{2mV_0}{\hbar^2}\right)^2}{4\left(\frac{2mE}{\hbar^2}\right)\left(\frac{2m(V_0 - E)}{\hbar^2}\right)} = \frac{V_0^2}{4E(V_0 - E)}$$

Substituting this back into the expression for $\left|\frac{A}{F}\right|^2$ gives

$$\left|\frac{A}{F}\right|^2 = 1 + \frac{V_0^2\sinh^2(\kappa L)}{4E(V_0 - E)}$$

Taking the reciprocal yields the exact expression for the transmission coefficient T:

$$T = \frac{1}{1 + \frac{V_0^2 \sinh^2(\kappa L)}{4E(V_0 - E)}}$$

Approximate Form for a Thick Barrier

For a wide or high potential barrier such that $\kappa L \gg 1$ (thick barrier approximation), the hyperbolic sine function can be approximated by its dominant exponential term:

$$\sinh(\kappa L) = \frac{e^{\kappa L} - e^{-\kappa L}}{2} \approx \frac{e^{\kappa L}}{2}$$

Squaring this expression yields

$$\sinh^2(\kappa L) \approx \frac{e^{2\kappa L}}{4}$$

Since $\kappa L \gg 1$, $e^{2\kappa L} \gg 1$, which makes the second term in the denominator of the exact transmission coefficient formula much larger than 1:

$$\frac{V_0^2 \sinh^2(\kappa L)}{4E(V_0 - E)} \gg 1$$

Neglecting the number 1 in the denominator of the transmission coefficient formula yields

$$T \approx \frac{1}{\frac{V_0^2}{4E(V_0 - E)} \left(\frac{e^{2\kappa L}}{4} \right)} = \frac{16E(V_0 - E)}{V_0^2} e^{-2\kappa L}$$

Because the exponential factor $e^{-2\kappa L}$ dominates the magnitude of T for large κL, the transmission coefficient is frequently written in the simplified exponential form:

$$T \approx e^{-2\kappa L}$$

where the decay parameter κ is defined by

$$\kappa = \frac{\sqrt{2m(V_0 - E)}}{\hbar}$$

Quantum Tunnelling Phenomenon and Physical Discussion

The non-zero transmission probability $T > 0$ for a particle with energy $E < V_0$ encountering a potential barrier of finite height V_0 and width L is known as **quantum tunnelling** (or the tunnel effect).

Classically, a particle cannot exist in a region where $E < V(x)$ because its kinetic energy $K = E - V(x) = \frac{p^2}{2m}$ would be negative, implying an imaginary momentum p, which is physically impossible in classical mechanics. Consequently, a classical particle hitting a barrier of height $V_0 > E$ is completely reflected with 100% probability at the boundary $x = 0$.

In quantum mechanics, the particle is described by a wave function. Inside the potential barrier (Region II), the Schrödinger equation leads to a solution involving decaying exponentials, $\psi_{II}(x) = Ce^{\kappa x} + De^{-\kappa x}$, rather than an oscillatory plane wave. Because the barrier width L is finite, the wave function decays exponentially across the barrier

but does not drop completely to zero at the far boundary $x = L$. As a result, the wave function connects smoothly to an oscillatory plane wave $\psi_{III}(x) = Fe^{ik_1 x}$ in Region III. The non-zero wave amplitude F past the barrier means there is a finite probability for the particle to appear on the far side of the barrier ($x > L$).

The transmission coefficient $T \approx e^{-2\kappa L}$ depends strongly on the physical parameters of the barrier and the particle:

1. **Barrier Width (L):** T decreases exponentially with increasing barrier width L. Doubling the width L drastically reduces the tunnelling probability by a factor of $e^{-2\kappa L}$.
2. **Barrier Height (V_0):** T decreases exponentially as the barrier height V_0 increases relative to the particle energy E. Higher values of $V_0 - E$ increase the attenuation factor $\kappa = \frac{\sqrt{2m(V_0 - E)}}{\hbar}$, causing the wave function to decay much faster inside the barrier.
3. **Particle Mass (m):** T decreases exponentially with increasing particle mass m. Because κ is proportional to $\sqrt{m}$, lighter particles such as electrons experience significantly higher tunnelling rates than heavier particles like protons or alpha particles under identical barrier conditions.

Questions

Short form first (2–3 marks), then long form (5–10 marks). **There are 4 Assam University questions for this unit in the paper bank**, from CHMHCC-502T, marked [AU] with their real source. The other questions are from other sources, marked as such.

Short-form questions

S1. [AU] Calculate the degree of degeneracy of a particle in a 3D cubical box of length a. — CHMHCC-502T, 5th Sem, 2020, CBCS (Sec A).

Answer in brief: Section 2 — $E = (h^2/8ma^2)(n_x^2 + n_y^2 + n_z^2)$; count the distinct (n_x, n_y, n_z) giving the same sum, e.g. (1,1,2) and its permutations are 3-fold degenerate, (1,1,1) is non-degenerate.

S2. [non-AU] Explain quantum tunnelling. — Univ. of Calicut, CBCSS B.Sc 5th Sem, PHY5 B07 Quantum Mechanics, Nov 2021, Q8 (2 marks).

Answer in brief: Section 4 — a particle of energy $E < V_0$ has a non-zero probability $T \approx e^{-2\kappa L}$ of appearing on the far side of a barrier it could not cross classically.

S3. [standard] Write the energy eigenvalues of a particle in a one-dimensional infinite potential well of width a.

Answer in brief: Section 1 — $E_n = n^2h^2/(8ma^2)$, $n = 1, 2, 3, \ldots$; the lowest energy is $E_1 = h^2/(8ma^2)$, not zero.

S4. [standard] Define the transmission and reflection coefficients for a particle meeting a step potential.

Answer in brief: Section 3 — R is the ratio of reflected to incident probability current and T the ratio of transmitted to incident probability current, with $R + T = 1$.

Long-form questions

L1. [AU] Solve Schrodinger's equation for a particle of mass m confined to move in a 1-D box of length a. — CHMHCC-502T, 5th Sem, 2020, CBCS (Sec B).

What to write: Section 1 in full — the Schrodinger equation inside the well, boundary conditions $\psi(0) = \psi(a) = 0$, $E_n = n^2h^2/(8ma^2)$ and $\psi_n(x) = \sqrt{2/a}\sin(n\pi x/a)$.

L2. [AU] Solve the Schrodinger equation for a particle in a 3-D box and find the energy. — CHMHCC-502T, 5th Sem, 2023, CBCS (Sec B).

What to write: Section 2 in full — separation of variables, $E = (h^2/8m)(n_x^2/a^2 + n_y^2/b^2 + n_z^2/c^2)$, normalized eigenfunction, and the cubical-box degeneracy table.

L3. [AU] Calculate the energy levels for the particle in a three-dimensional cubical box with edge ... Calculate the degeneracy ... — CHMHCC-502T, 5th Sem, 2021, CBCS (Sec B). Note: the edge length is garbled in the paper bank; use edge a.

What to write: Section 2 in full — $E = (h^2/8ma^2)(n_x^2 + n_y^2 + n_z^2)$; list the lowest levels with their degeneracies from the table.

L4. [non-AU] Obtain Schrodinger wave equation for a particle in one dimensional rigid box. Solve it to obtain normalized eigen functions and show that the eigen values are discrete. — Univ. of Calicut, CBCSS B.Sc 5th Sem, PHY5 B07 Quantum Mechanics, Nov 2021, Q21 (essay, 10 marks).

What to write: Section 1 in full — the boundary conditions allow only $k_n = n\pi/a$, so only the discrete energies $E_n = n^2h^2/(8ma^2)$ occur, with $\psi_n(x) = \sqrt{2/a}\sin(n\pi x/a)$.

L5. [standard] Calculate the transmission and reflection coefficients for a particle encountering a step potential, for E > V0 and E < V0, and discuss the results.

What to write: Section 3 in full — for $E > V_0$, $R = ((k_1 - k_2)/(k_1 + k_2))^2$ and $T = 4k_1k_2/(k_1 + k_2)^2$ with $R + T = 1$; for $E < V_0$, the wave decays exponentially and $R = 1, T = 0$; note the partial reflection even when $E > V_0$.

L6. [standard] Derive the transmission coefficient for a particle inside a finite rectangular potential barrier and explain the phenomenon of quantum tunnelling.

What to write: Section 4 in full — the three-region wave functions, matching at $x = 0$ and $x = L$, $T \approx e^{-2\kappa L}$ for a thick barrier with $\kappa = \sqrt{2m(V_0 - E)}/\hbar$, and why the particle can cross a classically forbidden region.

Sources

These notes were written from the unit syllabus only, as the user asked. No external PDF, textbook, or lecture notes were used. The questions are from the Assam University paper bank and other universities' papers (see the Questions section above), not from any source used for the notes.